

Exercise 2

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2.1:

The transit depth was calculated from the difference between the out-of-transit and in-transit flux levels.

$$\delta = F_{\text{out}} - F_{\text{in}}$$

$$\delta = 1.0000 - 0.9849$$

$$\delta = 0.0151$$

In parts per million:

$$\delta_{\text{ppm}} = 0.0151 \times 10^6$$

$$\delta_{\text{ppm}} = 15100 \text{ ppm}$$

The transit depth is therefore about 1.51%, which is much larger than an Earth-sized transit and is more consistent with a giant planet.

2.2:

For a simple transit model, the fractional loss of light is approximately equal to the area ratio of the planet and the star.

$$\delta = (R_p/R_*)^2$$

Therefore:

$$R_p/R_* = \sqrt{\delta}$$

$$R_p/R_* = \sqrt{0.0151}$$

$$R_p/R_* = 0.123$$

Thus, the planet radius is approximately 12.3% of the stellar radius.

2.3:

From the phase-folded light curve, the transit begins near phase -0.030 and ends near phase $+0.030$.

$$D_{\text{phase}} = 0.030 - (-0.030) = 0.060$$

The duration in days is:

$$D = D_{\text{phase}} \times P$$

$$D = 0.060 \times 2.18048 = 0.1308 \text{ days}$$

Converting into hours:

$$D = 0.1308 \times 24 = 3.14 \text{ hours}$$

Therefore, the transit duration is approximately 3.14 hours.

2.4:

Using the simplified transit-density relation:

$$\rho^* \approx 3P / (\pi^2 G D^3)$$

Convert the values into SI units:

$$P = 2.18048 \text{ days} = 188394 \text{ s}$$

$$D = 0.1308 \text{ days} = 11301 \text{ s}$$

$$G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$$

Substituting:

$$\rho^* \approx 5.94 \times 10^2 \text{ kg m}^{-3}$$

Relative to the Sun:

$$\rho^*/\rho_{\odot} = 594 / 1408$$

$$\rho^*/\rho_{\odot} \approx 0.42$$

The host star is therefore less dense than the Sun, suggesting a slightly inflated star.

2.5:

The effective temperature of the star is given as:

$$T_{\text{eff}} = 6099 \text{ K}$$

This is slightly hotter than the Sun, so the star is likely a late-F or early-G type main-sequence star. Using a main-sequence estimate:

$$R^* \approx 1.15 R_{\odot}$$

$$R^* = 1.15 \times 6.957 \times 10^8 \text{ m} = 8.00 \times 10^8 \text{ m}$$

2.6:

The stellar mass was estimated using:

$$\rho^* = M^* / [(4/3)\pi R^{*3}]$$

$$M^* = \rho^* \times (4/3)\pi R^{*3}$$

Substituting:

$$\rho^* = 5.94 \times 10^2 \text{ kg m}^{-3}$$

$$R^* = 8.00 \times 10^8 \text{ m}$$

$$M^* \approx 1.27 \times 10^{30} \text{ kg}$$

Relative to the Sun:

$$M^*/M_{\odot} = 1.27 \times 10^{30} / 1.989 \times 10^{30} \approx 0.64$$

The calculated stellar mass is somewhat lower than expected for a 6100 K star. This discrepancy most likely arises from approximations made during the transit-duration measurement and the assumption of zero limb darkening and perfectly central transits.

Final Results:

$$\text{Transit depth} = 0.0151$$

$$\text{Transit depth} = 15100 \text{ ppm}$$

$$R_p/R^* = 0.123$$

$$\text{Transit duration} = 3.14 \text{ hours}$$

$$\text{Stellar density} = 5.94 \times 10^2 \text{ kg m}^{-3}$$

$$\text{Stellar density} = 0.42 \rho_{\odot}$$

$$\text{Stellar radius} = 1.15 R_{\odot}$$

$$\text{Stellar mass} = 0.64 M_{\odot}$$

The T1 system shows the typical photometric signature of a short-period giant planet. Its deep transit, large radius ratio, and short orbital period are consistent with a hot-Jupiter like object. The derived stellar parameters are approximate because the simplified calculation neglects limb darkening, impact parameter effects, eccentricity, and blending. The strongest uncertainty comes from the manually measured transit duration, which directly affects the stellar density and mass estimates.

For T2:

2.1:

$$F_{\text{out}} = 1.0000$$

$$F_{\text{in}} \approx 0.9804$$

$$\delta = F_{\text{out}} - F_{\text{in}}$$

$$\delta = 1.0000 - 0.9804$$

$$\delta = 0.0196$$

$$\delta_{\text{ppm}} = 0.0196 \times 10^6 = 19600 \text{ ppm}$$

2.2:

$$\delta = (R_p/R_*)^2$$

$$R_p/R_* = \sqrt{0.0196} = 0.140$$

So, the planet radius is about 14.0% of the stellar radius.

2.3:

$$\text{Ingress phase} \approx -0.020$$

$$\text{Egress phase} \approx +0.020$$

$$D_{\text{phase}} = 0.040$$

$$D = D_{\text{phase}} \times P$$

$$D = 0.040 \times 5.441851$$

$$D = 0.2177 \text{ days} = 5.23 \text{ hours}$$

2.4:

$$\rho^* \approx 3P / (\pi^2 G D^3)$$

$$P = 5.441851 \text{ days} = 470176 \text{ s}$$

$$D = 0.2177 \text{ days} = 18807 \text{ s}$$

$$\rho^* \approx 3.22 \times 10^2 \text{ kg m}^{-3}$$

$$\rho^*/\rho_{\odot} \approx 322/1408 = 0.23$$

2.5:

$$T_{\text{eff}} \approx 6300 \text{ K}$$

Using the main-sequence temperature-radius diagram, a star near 6300 K corresponds to a radius of approximately:

$$R^* \approx 1.35 R_{\odot}$$

In SI units:

$$R^* = 1.35 \times 6.957 \times 10^8 \text{ m} = 9.39 \times 10^8 \text{ m}$$

2.6:

From Task 2.4:

$$\rho^*/\rho_{\odot} \approx 0.23$$

Using:

$$M^*/M_{\odot} = (\rho^*/\rho_{\odot})(R^*/R_{\odot})^3$$

$$M^*/M_{\odot} = 0.23 \times (1.35)^3 \approx 0.57$$

Therefore:

$$M^* \approx 0.57 M_{\odot}$$

The T2 system exhibits the characteristic features of a transiting giant exoplanet, including a deep transit profile, a large planet-to-star radius ratio, and a relatively long transit duration. The derived radius ratio of approximately 0.140 suggests a Jupiter-sized object orbiting a larger and hotter host star. From the temperature-radius relation, the host star is consistent with an F-type main-sequence star with a radius of about $1.35 R_{\odot}$. Although the stellar parameters remain approximate due to simplified assumptions and manual transit measurements, the overall system properties are physically consistent with a hot-Jupiter-like exoplanet orbiting a slightly evolved, low-density star.